

Logistic regression assumptions and diagnostics

Agenda and reminders

- + HW 2 due Friday
- + Department seminar tomorrow (9/11) at 11am in ZSR auditorium
 - + please refrain from wearing colognes, perfumes, and/or heavily scented body and hair products
- + Today: more on diagnostics

Shape assumption and quantile residuals

(In R)

Multicollinearity

What is multicollinearity?

Definition : one explanatory variable can be approximated by a linear combination of other explanatory variables in the data

Ex : $y_i \sim \text{Bernoulli}(\pi_i)$
$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$$

worst case : $x_{i1} = \alpha_2 x_{i2} + \alpha_3 x_{i3}$
$$\Rightarrow \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + (\beta_1 \alpha_2 + \beta_2) x_{i2} + (\beta_1 \alpha_3 + \beta_3) x_{i3}$$

Too many unknowns (β s), can't estimate coefficients
higher multicollinearity $\rightarrow$ more problems with estimation

Example: College scorecard data

The Department of Education compiles the College Scorecard, which is used to help prospective college students compare schools. For each school, variables include:

- + CONTROL: whether the school is public or private
- + SATVRMID: midpoint of SAT critical reading scores of students attending the school
- + ACTCMMID: midpoint of the ACT cumulative scores
- + UGDS: number of undergraduate students at the school
- + NPT4: average cost to attend the school
- + PCTFLOAN: fraction of undergraduates receiving a federal student loan
- + MD_EARN_WNE_P10: median salary of students 10 years after graduation

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Which of these variables may suffer from multicollinearity?

The impact of multicollinearity

How does correlation between the explanatory variables impact the fitted model?

- + Increased variability in coefficient estimates
- + Difficulty interpreting coefficients

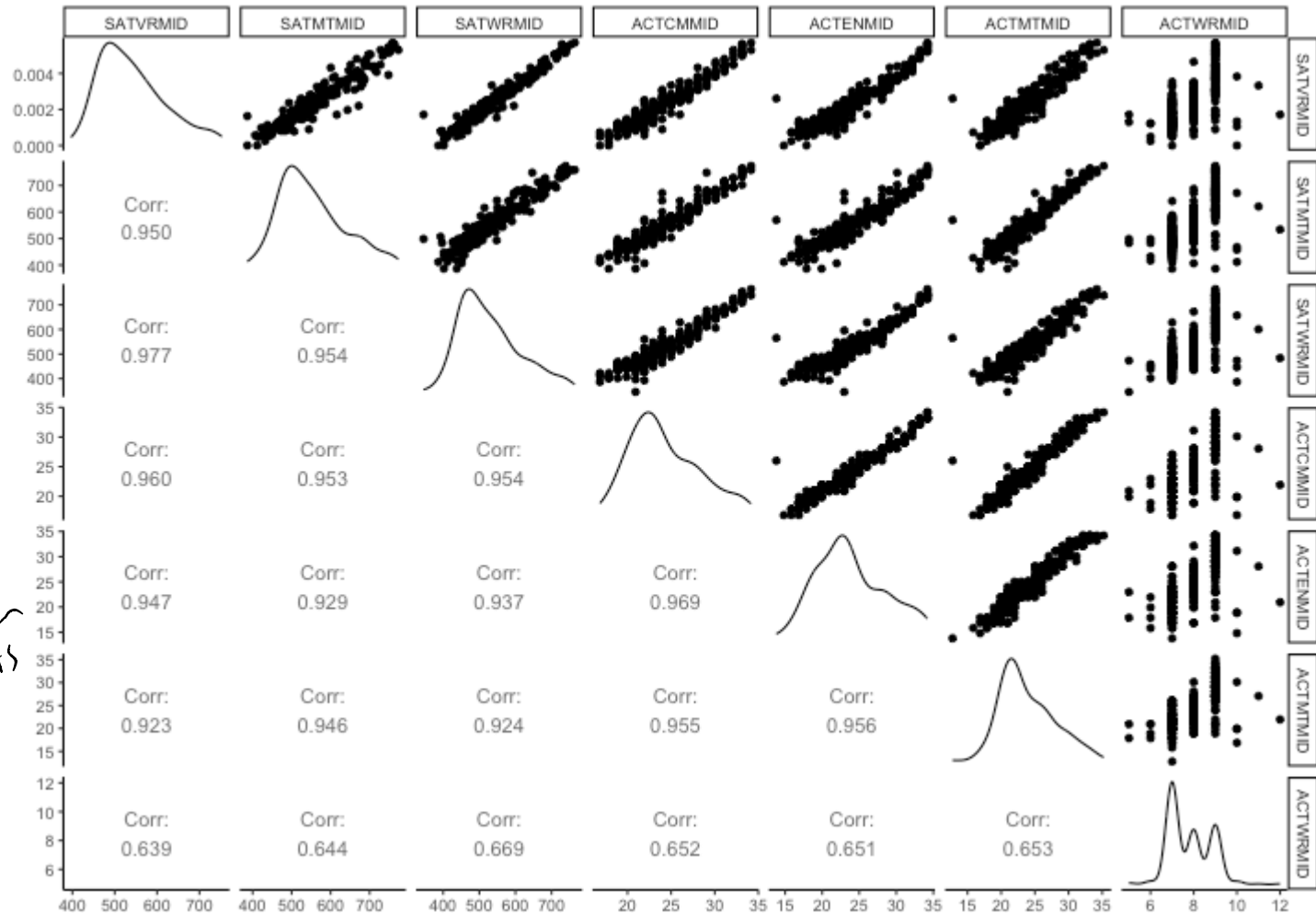
Diagnosing multicollinearity

How do I detect multicollinearity in my data?

- plot explanatory variables
- calculate correlation between explanatory variables
- variance inflation factors

Scatterplot and correlation matrix

Scatterplots for each pair of variables



correlation coefficients

Variance inflation factors

$$\text{VIF}_j = \frac{\text{Var}(\hat{\beta}_j) \text{ using all explanatory variables in model}}{\text{Var}(\hat{\beta}_j) \text{ using only } j^{\text{th}} \text{ explanatory variable}}$$

(j^{th} coefficient)

$$\text{VIF}_j \geq 1$$

$\text{VIF}_j \approx 1$ if variable j is unrelated to other explanatory variables

High VIF $\rightarrow$ high multicollinearity

Thresholds: usually concerned if $\text{VIF} > \text{threshold}$
(5 or 10)

Variance inflation factors in R

```
library(car)
```

```
m1 <- glm(RPY_3YR_70 ~ CONTROL + SATVRMID +  
          SATMTMID + SATWRMID + ACTCMMID + ACTENMID +  
          ACTMTMID + ACTWRMID + UGDS +  
          PCTFLOAN + MD_EARN_WNE_P10 + NPT4,  
          data = scorecard, family = binomial)  
vif(m1)
```

##	CONTROL	SATVRMID	SATMTMID	SATWRMID	ACTCMMID
##	3.511039	14.752767	11.246146	13.763868	11.6
##	ACTENMID	ACTMTMID	ACTWRMID	UGDS	PCTFLOAN
##	12.258720	8.837329	1.671265	2.426494	2.6
##	MD_EARN_WNE_P10	NPT4			
##	1.333261	2.536284			

Almost all test score variables
have high multicollinearity!

Addressing model issues

How should we handle multicollinearity in the explanatory variables? Discuss with a neighbor for a few minutes, then we will discuss as a group.

- combine variables (e.g., overall test score variable)
- fit different models for different variables
- remove some variables (e.g., choose one test score to represent all test scores)

Class activity

https://sta214-f25.github.io/class_activities/ca_08.html

At the end of class, render to HTML and submit HTML on Canvas